MARK SCHEME · TEACHER COPY

10BR · Chapter 6: Automated and Emerging Technologies

Cambridge IGCSE Computer Science 0478

Paper 1 Computer Systems

Total marks: 33

Official Cambridge mark scheme extracts for each question in the matching exam paper, with question numbers changed to match the exam. Source of each question is shown above its rows.

For teacher / admin use only. Do not give to students before they have attempted the paper.

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 Total: 33 marks

Question 1 · May/June 2024 Paper 12, original Q10 (a),(b) [7 marks]

Question	Answer	Marks
1	0 0 1 0 1 1 0	
1(a)	Any **two** from: • The **simulation** of intelligent behaviours by computers • A collection of data and the rules for using that data • Has the ability to reason • Has the ability to learn/adapt	**2**
1(b)	Any **five** from: • The user will enter data into the **interface** // The user will plug the car into an **interface** // The user is given questions using the **interface** • The **inference engine** will decide which questions to ask • … by using the previous answers given • The **inference engine** will decide on a diagnosis • … by **comparing the data/answers** entered to the **knowledge base** and **rule base** • … by calculating which option is most likely if there are multiple • The **interface** will output the **diagnosis/result** • The **explanation system** shows **how** the diagnosis was reached	**5**

Question 2 · May/June 2025 Paper 12, original Q5 (a),(b) [11 marks]

Question	Answer	Marks
2(a)	**Three** from: • The inference engine is used … • … to **decide** which questions to ask the user … • … based on the **previous** data input. • Symptoms input are located in/compared to knowledge base. • … then applies the rule base to the knowledge base (to decide the diagnosis)	**3**

2(b)(i)	Any **two** from: Examples: • Sensors • Microprocessors • Actuators	**2**
2(b)(ii)	Any **four** from: Examples: • a doctor doesn't need to **travel** to the hospital to do the surgery. • … so it can be done by any specialist/doctor in the world. • … so it can be done immediately without needing to wait for travel time // reduces the waiting time for the patient. • … so the doctor may be better at the surgery as they won't be tired from travel. • … so travel costs are saved. • Surgery with robots enhances precision/accuracy. • … so a smaller incision can be made. • … so the recovery time may be shorter. • … as the components used to enter the body can be much smaller than a human hand. • … so the surgery is safer/ more hygienic // by example e.g. can stop the doctor needing to be near an infectious patient. • The surgery may have a higher rate of success	**4**

2(b)(iii)	**Two** from (one for a point and one for the matching expansion): Examples: • The internet connection could be lost/delayed … • … so the surgery may not be able to continue. • The robot will be expensive to buy/maintain … • … this money could have been spent on other causes. • The robot could be hacked …. • … and endanger the patient's life. • Data could be corrupted in transmission … • … changing the nature of the instruction for the robot. • The robot's hardware could malfunction … • … so the surgery cannot continue	**2**

Question 3 · May/June 2026 Paper 11, original Q7 (a),(b),(c),(d) [11 marks]

Question	Answer	Marks	Guidance
3(a)	Acoustic sensor	**1**	Mark the first answer given. Sound/noise sensor is TV. Microphone is NE.
3(b)	Any **two** from: • Proximity sensor • Infrared sensor • Light sensor • Magnetic field sensor	**2**	Mark the first answer given.
3(c)(i)	Any **one** from: • It has a mechanical structure/framework • It is programmable • It has the ability to move	**1**	Mark whole answer. MP1 – It is mechanical is TV MP2 – It can follow instructions is TV

3(c)(ii)	Any **three** from: • The microprocessor/it compares the received data/data sent to the stored value/preset value (of 10) • If the **data/value** received is **less than 10**, (the microprocessor) sends a **signal** to stop the robot • (The signal is sent) to an actuator that is used to stop the robot	**3**	MP2 – 'when an object/distance/robot is less than 10cm' is TV. MP3 – just reference to an actuator is NE here. Must be saying the actuator is what is used to trigger the action to stop the robot in some way.
3(d)	Any **four** from: For example: • It can use machine learning • … by adapting its own data/rules/processes • It can **create/makes** a (digital) map (of the room) // Records/collects data about the **shape/dimensions** of the room • … and **record/store** the **location** of objects • **GPS** may be used to record the location of objects • It can **update** its (digital) map • … to **record/store** when objects are **moved/removed/added** • It can store successful/unsuccessful routes/actions (that avoid objects)	**4**	Do not accept 'it can learn where objects are placed in the room'. This is given in the question.

Question 4 · May/June 2026 Paper 12, original Q6 (b) [4 marks]

Question	Answer	Marks	Guidance
4	Examiners must award marks according to the mark scheme and **must not** award marks with grade thresholds or grade descriptions in mind.		

Annotation	Meaning
✓	Correct point
✗	Incorrect point
?	Unclear response
FT	Follow through
REP	Repetition

I	Ignore
BOD	Benefit of doubt given
NBOD	Benefit of doubt not given
TV	Content of response too vague
NAQ	Not answered question
Annotation	**Meaning**
λ	Omission
	Section not relevant
	Section incorrect
Highlighter	Information copied from the text
SEEN	Page or response seen by examiner

4(a)	**Four** from: For example: • It can use machine learning • … by **adapting** its **own data/rules/processes** • It can use image recognition • It can **analyse/identify** the **patterns** (in different flowers) • It can use **feedback from user/human** • … to **identify/store successful/unsuccessful** identifications • It can be **trained** by being shown pictures of flowers … • … to learn in more detail/new characteristics of flowers	**4**	MP2 – Just 'it can learn and adapt' is NE. MP3 - Accept other references outside the syllabus to AI that could be used to improve e.g. neural networks. MP5 – 'learn from past results/mistakes' is TV. MP5 – BOD reference to feedback being manual in place of 'from user'